

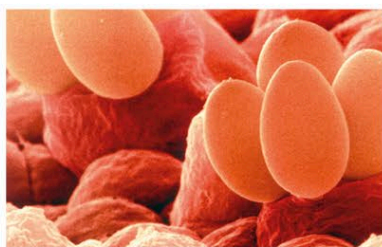
MUSHROOMS

The phylum Basidiomycota includes the majority of what are commonly called mushrooms and toadstools. Found in almost every major habitat type, nearly all share the ability to form sexual spores externally on special cells called basidia.

PHYLUM	BASIDIOMYCOTA
CLASSES	16
ORDERS	52
FAMILIES	177
SPECIES	About 32,000



These typical mushrooms show the cap with radiating gills on the underside, supported by the central stem.



The microscopic spores of the field mushroom are here shown still attached to the basidia, or reproductive cells.



This stinkhorn is covered with a dark spore mass, which rapidly liquifies and emits a foul odour, usually of rotting flesh.

DEBATE

COLOURFUL MYSTERY

Many mushrooms have bright colours: red, violet, blue, and green. Scientists have little idea what advantage these colours give to the mushroom. Unlike flowers, they do not need bright colours to attract pollinators. Red and orange pigments possibly prevent damage from sunlight, while others may act as a warning to predators.

Few fungi are as diverse in form as the Basidiomycota. Their fruitbodies (the structures in which spores are produced) have evolved to provide varied and efficient mechanisms for spore dispersal. These fruitbodies often take the form of a stem, cap, and gills, but many are simple, crust-like sheets, and others are more complex, bracket-like forms. More exotic still are the totally enclosed, spherical types such as the puffballs and the earthstars. Then there are the beautiful, coral-like structures of the club and coral fungi, and the often weird, animal-like shapes of the stinkhorns. Each structure supports and produces special spore-producing cells called basidia, and – depending on the fungus concerned – uses a variety of mechanisms to disperse these spores. For example, gilled fungi forcibly eject their spores, to be carried off by air currents. The puffballs, however, depend on a combination of wind and raindrops to puff out their spores. Stinkhorns, on the other hand, use foul odours and bright colours to attract animals, such as insects and other invertebrates, to eat their spore mass and to pass the spores unharmed through their digestive tracts. In this way the spores are scattered as far as the animals travel before excreting them.

FUNGAL FEEDING

The main body of the fungus is usually underground, formed of fine fungal threads called hyphae, which make up a mycelium (fungal body). These threads permeate the substance on which the fungus grows, be it soil, old leaf litter, fallen timber, living plant tissues, or even dead and decaying animals. The mycelium may spread over a large area, and in many fungi it also forms a mutually beneficial relationship with the roots of plants. In this relationship – called a mycorrhiza – the mycelium wraps around and penetrates the plant's roots, providing the fungus with access to carbohydrates from the plant. In return, the plant gains from the mycelium's greater ability to absorb water and mineral nutrients. Other mushrooms break down dead organic matter, or devour living organisms for food – both processes help to renew the soil and to improve the growing conditions for other living things.

MUSHROOMS AMONG FALLEN LEAVES >

Many fungi feed on dead and decaying plant tissues and may spread in huge numbers where nutrients and moisture occur.

